

Prompt Engineering and Retrieval Augmented Generation

Agenda

- Retrieval Augmented Augmentation (RAG) Quiz
- Prompt Engineering
- Building Blocks of RAG
- Chunking
- Semantic Search

Let's begin the discussion by answering a few questions.

Prompt Engineering and RAG Quiz

What does "constrained outputs" refer to in the context of prompt engineering?

A

Unrestricted and free-form model responses

B

Limiting the allowable model responses

C

Model training without any guidelines

D

Enhancing model creativity

Prompt Engineering and RAG Quiz

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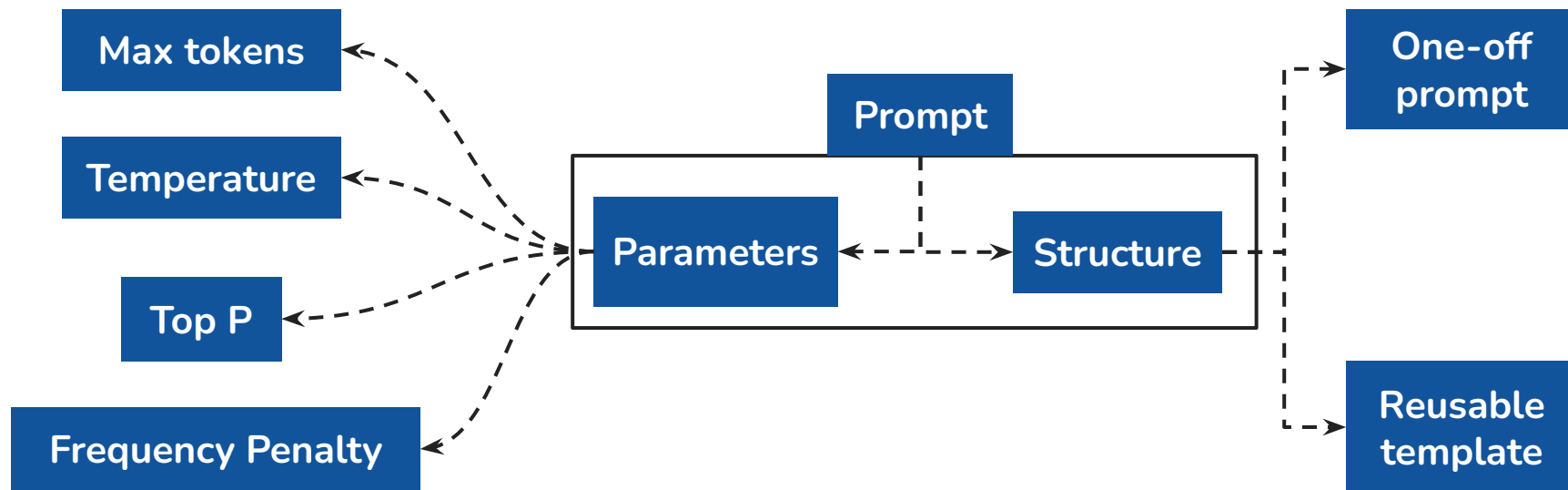
D

Enhancing model creativity

Prompt Engineering

Prompt: Specific set of instructions sent to a LLM to accomplish a task

Engineering: Iteratively deriving a specific prompt for the task



Significance of Prompt Engineering



PROMPT

Write a blog post about the benefits of meditation for mental health, targeting a beginner audience and optimized for SEO

1

Control model behaviour

The prompt is designed to control the behavior of the LLM by specifying the topic, target audience, and optimization criteria.

2

Get constrained outputs

The prompt includes specific keywords and phrases, such as "meditation," "mental health," "beginner," and "SEO," which constrain the model's response to a particular domain and style.

3

Higher output quality

The prompt provides context and specifies the tone or style, which helps the model understand the task at hand and generate a high-quality response.

4

Automate LLM operations

The prompt can be used to automate various LLM operations, such as fine-tuning, evaluation, and deployment.

Prompt Engineering and RAG Quiz

Which of the following represents the correct sequence of blocks for processing data in a retrieval-augmented generation (RAG) model?

A

Embedding -> Vector DB -> Data Chunk -> Retriever -> LLM

B

Data Chunk -> Embedding -> Vector DB -> Retriever -> LLM

C

Vector DB -> Data Chunk -> Retriever -> Embedding -> LLM

D

Data Chunk -> Embedding -> Retriever -> Vector DB

Prompt Engineering and RAG Quiz

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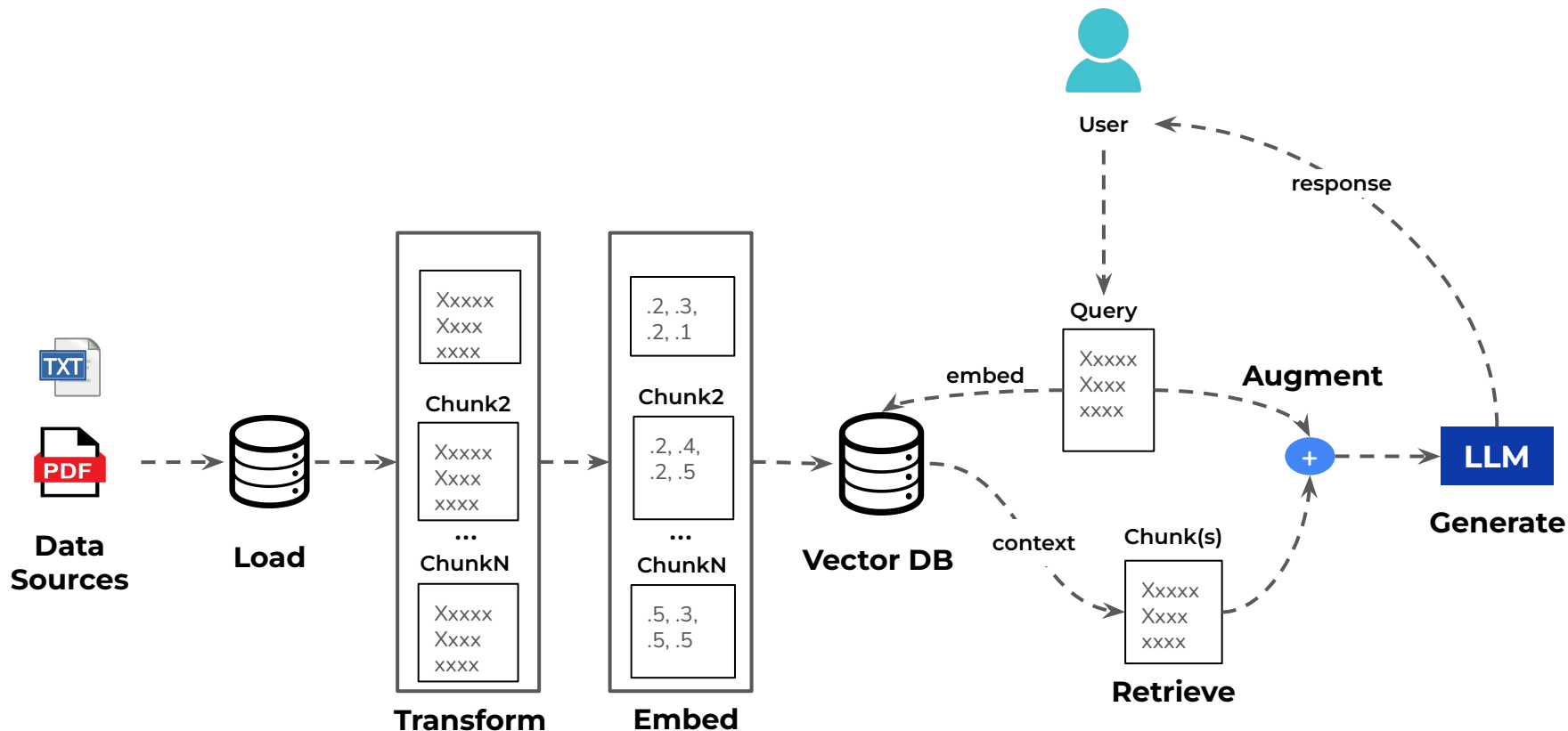
C

Vector DB -> Data Chunk -> Retriever -> Embedding -> LLM

D

Data Chunk -> Embedding -> Retriever -> Vector DB

Retrieval Augmented Generation (RAG)



Prompt Engineering and RAG Quiz

How does increasing the chunk overlap in a RAG system impact retrieval?

A

It reduces the number of retrieved chunks, improving efficiency

B

It increases redundancy, but helps maintain continuity in retrieved content

C

It eliminates the need for chunking altogether

D

It prevents the model from retrieving any irrelevant information

Prompt Engineering and RAG Quiz

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Chunking

The process of **breaking a document** (or text corpus) into **smaller, manageable pieces** (chunks) before storing them

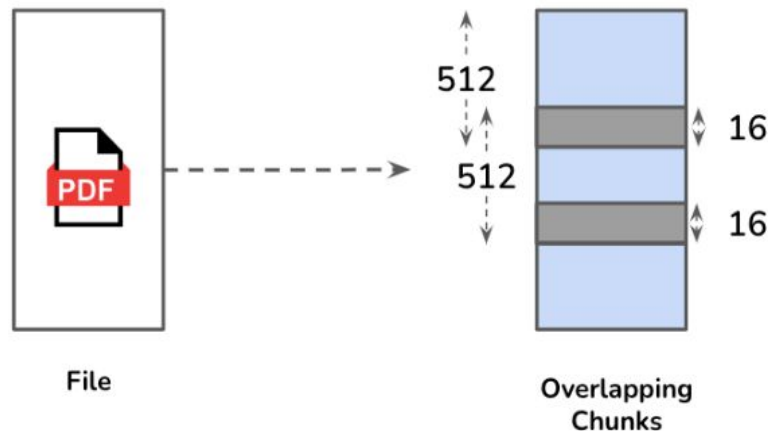
Chunks are then **indexed and used to retrieve relevant information efficiently** when generating responses

512 => number of tokens in a chunk

16 => number of tokens overlapping between any two consecutive chunks

Increasing the chunk overlap parameter increases redundancy as there will be larger overlaps

But this helps maintain continuity in retrieved content as information might be spread across multiple chunks



Prompt Engineering and RAG Quiz

Why do we need to create embeddings for our data in a Retrieval-Augmented Generation (RAG) system?

- A To improve keyword-based searching
- B To enable semantic similarity search
- C To reduce the amount of stored data
- D To make data more structured

Prompt Engineering and RAG Quiz

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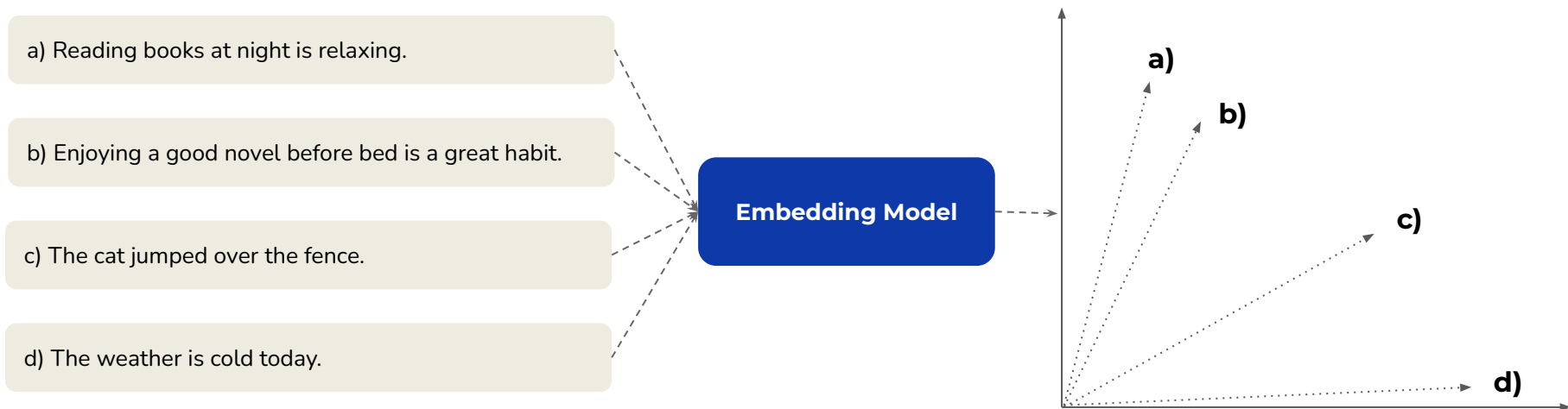
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To reduce the amount of stored data

D

To make data more structured

Sentence Embeddings



Similar sentences are closer in the embedding space; dissimilar sentences are further apart

This enables **semantic search to retrieve sentences** (or chunks) from the data source that are **“similar” to the user query**

Prompt Engineering and RAG Quiz

How does RAG enhance the capabilities of LLMs in natural language processing tasks ?

A

By limiting the access to external knowledge sources for generating responses

B

By reducing the need for fine-tuning LLMs on specific tasks

C

By integrating external knowledge sources to provide contextually rich and accurate responses

D

By reducing the model's size

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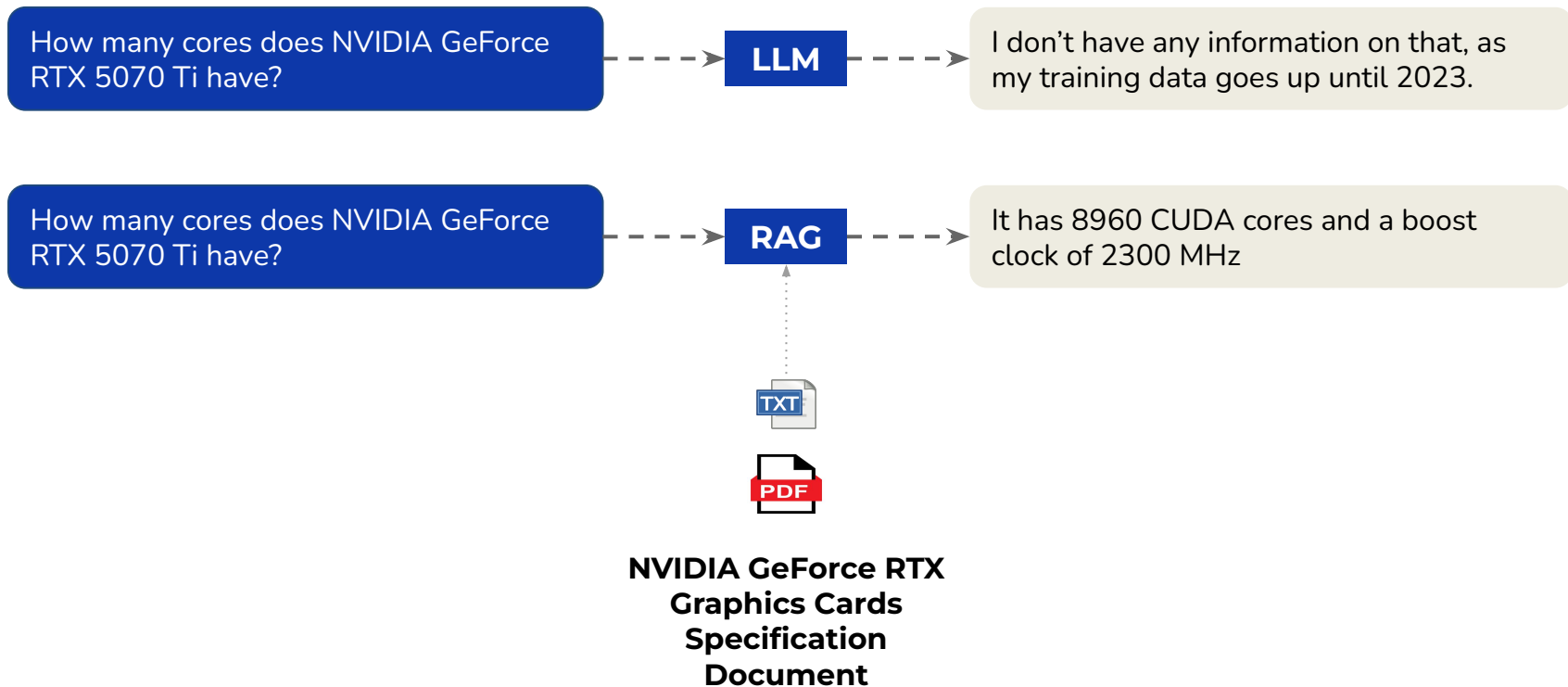
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Advantage of RAG





Power Ahead!

